

PREPARE-THRIVE-SUCCEED

IPU NOTES

B.TECH CSE

2nd Year



SNAPED

COMMUNITY

CM Unit-4

to read today

JOIN US:



Solu. of Ordinary differential Equations

① Picards Method of successive Approximations

(a) To solve $\frac{dy}{dx} = f(x, y)$; $y(x_0) = y_0$

First approximation $y^{(1)} = y_0 + \int_{x_0}^x f(x, y) dx$

Second approx. $y^{(2)} = y_0 + \int_{x_0}^x f(x, y^{(1)}) dx$

n^{th} approx. $y^{(n)} = y_0 + \int_{x_0}^x f(x, y^{(n-1)}) dx$

The process is stopped when the two values of y viz. $y^{(n-1)}$ and $y^{(n)}$ are same to the desired degree of accuracy.

(b) To solve the simultaneous equations

$$\frac{dy}{dx} = \phi(x, y, z) \quad \text{and} \quad \frac{dz}{dx} = f(x, y, z)$$

with initial conditions $y(x_0) = y_0$, $z(x_0) = z_0$

First approx. $y^{(1)} = y_0 + \int_{x_0}^x \phi(x, y_0, z_0) dx$; $z^{(1)} = z_0 + \int_{x_0}^x f(x, y_0, z_0) dx$

2nd approx. $y^{(2)} = y_0 + \int_{x_0}^x \phi(x, y^{(1)}, z^{(1)}) dx$, $z^{(2)} = z_0 + \int_{x_0}^x f(x, y^{(1)}, z^{(1)}) dx$

etc.

(c) To solve the diff eqn $\frac{dy}{dx} = f(x, y, \frac{dy}{dx})$

Putting $\frac{dy}{dx} = z$, above eqn can be written as first order simultaneous equations

$$\frac{dy}{dx} = z, \quad \frac{dz}{dx} = f(x, y, z)$$

and can be solved as in (b)

Ex.1 Employ Picard's method to obtain the solution of

$$\frac{dy}{dx} = x^2 + y^2 \quad \text{for } x \neq 0.4$$

correct to four decimal places given that $y \neq 0$ when $x = 0$

Solu:-

$$f(x, y) = x^2 + y^2, \quad y(0) = 0.$$

$$\text{First approx. } y^{(1)} = 0 + \int_0^x (x^2 + 0) dx = \frac{x^3}{3}$$

$$y^{(1)}(0.4) = .02133$$

$$y^{(2)} = 0 + \int_0^x (x^2 + \frac{x^6}{9}) dx = \frac{x^3}{3} + \frac{x^7}{63}$$

$$y^{(2)}(0.4) = .02136$$

~~$$y^{(3)} = 0 + \int_0^x x^2 + \dots$$~~

Thus $y^{(1)}$ & $y^{(2)}$ are same .0213 upto 4th place of decimal

$\therefore$ Repd. ans is .0213

Ex.2 Using Picard's method, solve the equations

$$\frac{dy}{dx} = x + z, \quad \frac{dz}{dx} = x - y^2$$

with initial conditions $y(0) = 2, z(0) = 1$

Obtain the values of y and z from II approximation when $x = .2$

Solu: $\phi(x, y, z) = x + z, \quad f(x, y, z) = x - y^2$

$$x_0 = 0, \quad y_0 = 2, \quad z_0 = 1$$

$$\text{First approx. } y^{(1)} = y_0 + \int_{x_0}^x \phi(x, y_0, z_0) dx, \quad z^{(1)} = z_0 + \int_{x_0}^x f(x, y_0, z_0) dx$$

$$y^{(1)} = 2 + \int_0^x (x + 1) dx = 2x + \frac{x^2}{2}, \quad z^{(1)} = 1 + \int_0^x (x - 4) dx = 1 - 4x + \frac{x^2}{2}$$

$$\text{Second approx. } y^{(2)} = 2 + \int_0^x (x + 1 - 4x + \frac{x^2}{2}) dx = 2 + x - \frac{3x^2}{2} + \frac{x^3}{6}$$

$$z^{(2)} = 1 + \int_0^x \left\{ x - \left(4 + x^2 + \frac{x^4}{4} + 4x + x^3 + 2x^2 \right) \right\} dx$$

$$= 1 - 4x - \frac{3x^2}{2} - x^3 - \frac{x^4}{4} - \frac{x^5}{5}$$

$$y(1.2) = 2.14133 \text{ approx.}$$

$$z(1.2) = 0.13158 \text{ approx.}$$

Ex.3. Solve by Picard's method, the differential equation

$$\frac{d^2 y}{dx^2} = x^3 \left(\frac{dy}{dx} + y \right)$$

when $y=1$, $\frac{dy}{dx} = \frac{1}{2}$ when $x=0$, obtain result upto ~~III~~ approximation and calculate y when $x = 0.1$ approximately.

Soln.: - Given $\frac{d^2 y}{dx^2} = x^3 \left(\frac{dy}{dx} + y \right)$

Putting $\frac{dy}{dx} = z$, we get $y_0 = 1, z_0 = \frac{1}{2}$

$$\frac{dz}{dx} = x^3 (y + z)$$

First approximation $y^{(1)} = y_0 + \int_0^x f(x, y_0, z_0) dx$

$$= 1 + \int_0^x \frac{1}{2} dx = 1 + \frac{x}{2}$$

Second approximation: $z^{(1)} = z_0 + \int_0^x f(x, y_0, z_0) dx = \frac{1}{2} + \int_0^x [x^3 (1 + \frac{1}{2}) + \frac{1}{2}] dx$

$$= 1 + x + \frac{x^2}{2}$$

$$z^{(1)} = \frac{1}{2} + \frac{3x^4}{4} + \frac{5}{10} + \frac{1}{2}x$$

$$y^{(2)} = 1 + \int_0^x \left(\frac{1}{2} + \frac{3}{8}x^4 \right) dx$$

$$= 1 + \frac{1}{2}x + \frac{3}{40}x^5$$

$$z^{(2)} = \frac{1}{2} + \int_0^x x^3 \left(1 + \frac{x}{2} + \frac{1}{2} + \frac{3}{8}x^4 \right) dx$$

$$= \frac{1}{2} + \frac{3}{8}x^4 + \frac{x^5}{10} + \frac{3x^8}{64}$$

$$\therefore y(0.1) = 1.05 \quad \underline{\underline{\text{Ans.}}}$$

~~Q~~

② Taylor's series method :

Let the first order diff. eqn. be

$$\frac{dy}{dx} = f(x, y) \quad \text{--- ①}$$

with initial condition $y(x_0) = y_0$

Diff ① $\frac{dy}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx}$

i.e. $y'' = f_x + f_y y'$ --- ②

Diff this successively, we can get $y''', y^{(4)}$ etc.

Using initial condns we can get $(y')_0, (y'')_0, (y''')_0$ etc.

Hence the Taylor series

$$y = y_0 + (x-x_0)(y')_0 + \frac{(x-x_0)^2}{2!}(y'')_0 + \frac{(x-x_0)^3}{3!}(y''')_0 + \dots \quad \text{--- ③}$$

gives the values of y for any value of x for which ③ converges.

Ex. 1. Obtain Picard's second approximate value of y at $x=1$ when $x=0$ soln. of the initial value problem $y' = \frac{x^2}{y^2+1}, y(0)=0$

Soln: We know that by Picard's rule

$$y_1 = y_0 + \int_{x_0}^x f(x, y_0) dx$$

Here $y_0 = 0, x_0 = 0, f(x, y) = \frac{x^2}{y^2+1}$

$$\therefore y_1 = 0 + \int_0^x \frac{x^2}{0^2+1} dx = \int_0^x x^2 dx = \frac{x^3}{3}$$

2nd Approx. $y_2 = y_0 + \int_0^x \frac{x^2}{\frac{x^6}{9}+1} dx = 0 + \int_0^x \frac{x^2}{\frac{x^6}{9}+1} dx$

$$= 9 \cdot \left[\frac{1}{3} \tan^{-1} \frac{x}{3} \right]_0^x = \tan^{-1} \frac{x}{3}$$

$x^3 = t$
 $3x^2 dx = dt$
 $\int \frac{dx}{t^2+9}$

Q.2. Find an approximate value of y when $x=0.1$ if $dy/dx = x-y^2$ & $y=1$ at $x=0$. using
(a) Picard's method (b) Taylor's series

Sol. Picard's Method:

By Picard's method 1st approximation y_1 is given by

$$y_1 = y_0 + \int_{x_0}^x f(x, y_0) dx.$$

Here $y_0 = 1$, $x_0 = 0$, $f(x, y) = x - y^2$

$$\therefore y_1 = 1 + \int_0^x (x-1) dx = 1 + \frac{x^2}{2} - x$$

$$y_2 = y_0 + \int_{x_0}^x f(x, y_1) dx$$

$$= 1 + \int_0^x \left[x - \left(1 + \frac{x^2}{2} - x \right)^2 \right] dx$$

$$= 1 + \int_0^x \left[x - \left\{ 1 + \frac{x^4}{4} + x^2 + x^2 + 2x + \frac{x^3}{2} \right\} \right] dx$$

$$= 1 + \int_0^x \left[1 + 3x - 2x^2 + \frac{x^3}{2} - \frac{x^4}{4} \right] dx$$

$$= 1 + x + \frac{3}{2}x^2 - \frac{2}{3}x^3 + \frac{x^4}{8} - \frac{x^5}{20}$$

$\therefore y$ at $x=0.1$ is given by

$$y = 1 + \frac{3}{2}(.1)^2 - \frac{2}{3}(.1)^3 + \frac{(.1)^4}{8} - \frac{(.1)^5}{20}$$

$$= 1 + .15 - .0007 + .00001 -$$

$$= 0.9143 \quad \underline{\text{Ans.}}$$

Taylor's Method:

$$\frac{dy}{dx} = x - y^2 \quad \therefore (y')_0 = 0 - 1 = -1$$

$$\therefore \frac{d^2y}{dx^2} = 1 + (-2y) \cdot \left(\frac{dy}{dx} \right)$$

$$= 1 - 2y \frac{dy}{dx}$$

$$y''_0 = 1 - 2 \cdot 1 \cdot (-1) = 3$$

$$\frac{d^3y}{dx^3} = 0 - 2 \left[\frac{dy}{dx} \cdot \frac{dy}{dx} + y \cdot \frac{d^2y}{dx^2} \right] \quad \therefore (y''')_0 = -2[1 + 1 \cdot 3]$$

$$= -2[4]$$

$$= -8$$

$$\frac{dy}{dx} = -2 \left[\begin{array}{l} \end{array} \right]$$

$$\therefore y = y_0 + (x-x_0)(y')_0 + \frac{(x-x_0)^2}{2!}(y'')_0 + \frac{(x-x_0)^3}{3!}(y''')_0 + \dots$$

$$\therefore y = 1 + (x-0)(-1) + \frac{x^2}{2!} \cdot 3 + \frac{x^3}{3!} \cdot 4 + \dots$$

$$(y)_{.1} = 1 + (-1) \times (-1) + \frac{.01}{2} \cdot 3 + \frac{-.001}{6} \times (-8) + \dots$$

$$= 1 - 1 + .015 + .0013 + \dots$$

$$= .9147$$

② Euler's method:

To solve $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$

General formula is

$$y_{n+1} = y_n + h f(x_n, y_n)$$

(b) Improved Euler's method:

$$y_1^{(n+1)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(n)})] \quad n=0, 1, 2, \dots$$

where $y_1^{(n)}$ is the n^{th} approximation to y_1

$$\text{we choose } y_1^{(1)} = y_0 + h f(x_0, y_0)$$

(c) Modified Euler's method:

$$y_{n+1} = y_n + h f\left\{x_n + \frac{h}{2}, y_n + \frac{h}{2} f(x_n, y_n)\right\}$$

Ex.1. Solve $\frac{dy}{dx} = x^2 + y$, $y(0) = 1$ to obtain
 $y(0.02)$ and $y(0.04)$ using (i) Euler's method
(ii) Euler's improved method & (iii) Euler's
modified method.

Soln: (i) Euler's method: $x_0 = 0$, $y_0 = 1$
Taking $h = 0.02$,

$$y_1 = y_0 + h f(x_0, y_0) = y_0 + h (x_0^2 + y_0) = 1 + 0.02 \times 1 = 1.02$$

$$\therefore \boxed{y(0.02) = 1.02}$$

$$\begin{aligned} y_2 &= y_1 + h f(x_1, y_1) = 1.02 + 0.02 (0.0004 + 1.02) \\ &= 1.02 + 0.02 (1.0204) \\ &= 1.040408 \end{aligned}$$

$$\boxed{y(0.04) = 1.040408}$$

(ii) Improved Euler's method:

$$y_1^{(1)} = 1.02.$$

$$y_1^{(2)} = y_0 + \frac{.02}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})]$$

$$= 1 + .01 [1 + .0004 + 1.02]$$

$$= 1 + .01 [2.0204] = 1 + .020204$$

$$= 1.020204$$

$$y_1^{(3)} = 1 + .01 [1 + .0004 + 1.020204] = 1 + .01 [2.020604]$$

$$= 1.02020604$$

$$\therefore y_1 = \boxed{y(1.02) = 1.0202} \text{ upto four decimal places}$$

$$y_2^{(1)} = y_2 = y_1 + h f(x_1, y_1) = 1.0202 + .02 (.0004 + 1.0202)$$

$$= 1.0202 + .02 (1.0206)$$

$$= 1.040612$$

$$y_2^{(2)} = y_1 + \frac{.02}{2} [f(x_1, y_1) + f(x_2, y_2^{(1)})]$$

$$= y_1 + .01 [x_1^2 + y_1 + x_2^2 + y_2^{(1)}]$$

$$= 1.0202 + .01 [.0004 + 1.0202 + .0016 + 1.040612]$$

$$= 1.0202 + .01 [2.062812] = 1.040828$$

$$y_2^{(3)} = 1.0202 + .01 [.0004 + 1.0202 + .0016 + 1.040828]$$

$$= 1.0202 + .01 [2.063028]$$

$$= 1.040830$$

$$\therefore y_2 = \boxed{y(1.04) = 1.0408} \text{ upto 4 decimal places}$$

(iii) Euler's modified method:

$$\begin{aligned}y_1 &= y_0 + h f\left\{x_0 + \frac{h}{2}, y_0 + \frac{h}{2} f(x_0, y_0)\right\} \\&= y_0 + h \left\{ \left(x_0 + \frac{h}{2}\right)^2 + y_0 + \frac{h}{2} (x_0^2 + y_0) \right\} \\&= 1 + .02 \{ .0001 + 1 + .01 \times 1 \} = 1 + .02 \times 2.0101 \\&= 1.020202\end{aligned}$$

i.e. $y(1.02) = 1.0202$ upto four decimal places.

$$\begin{aligned}y_2 &= y_1 + h \left\{ \left(x_1 + \frac{h}{2}\right)^2 + y_1 + \frac{h}{2} (x_1^2 + y_1) \right\} \\&= 1.0202 + .02 \{ (.02 + .01)^2 + 1.0202 + .01 (.0004 + 1.0202) \} \\&= 1.0202 + .02 \{ .0009 + 1.0202 + .010206 \} \\&= 1.0202 + .02 \{ 1.031306 \} \\&= 1.04082612\end{aligned}$$

$\therefore y(1.04) = 1.0408$ upto four decimal places.

Single Step Methods

1. Runge-Kutta second order method

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

$$y_{i+1} = y_i + \frac{1}{2} (k_1 + k_2) \text{ where } k_1 = hf(x_i, y_i)$$

$$k_2 = hf(x_i + h, y_i + k_1)$$

We start from $y_1 = y_0 + \frac{1}{2} (k_1 + k_2)$

where $k_1 = hf(x_0, y_0)$, $k_2 = hf(x_0 + h, y_0 + k_1)$

2. Runge-Kutta third order method :

Euler-like
or Heun-like

$$y_{i+1} = y_i + \frac{1}{6} (k_1 + 4k_2 + k_3)$$

where $k_1 = hf(x_i, y_i)$

$$k_2 = hf(x_i + \frac{h}{2}, y_i + \frac{k_1}{2})$$

$$k_3 = hf(x_i + h, y_i + k_1) \text{ where } k_1' = hf(x_i + h, y_i + k_1)$$

$$y_{i+1} = y_i + \frac{1}{3} (2k_1 + 3k_2 + k_3)$$

$$k_1 = hf(x_i, y_i)$$

$$k_2 = hf(x_i + \frac{2h}{3}, y_i + \frac{2k_1}{3})$$

$$k_3 = hf(x_i + h, y_i + k_1)$$

Start from $y_1 = y_0 + \frac{1}{6} (k_1 + 4k_2 + k_3)$

with $k_1 = hf(x_0, y_0)$, $k_2 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2})$

$$k_3 = hf(x_0 + h, y_0 + k_1) \text{ \& } k_1' = hf(x_0 + h, y_0 + k_1)$$

3. RK fourth order or classical RK method :

$$y_{i+1} = y_i + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

where $k_1 = hf(x_i, y_i)$, $k_2 = hf(x_i + \frac{h}{2}, y_i + \frac{1}{2} k_1)$

$$k_3 = hf(x_i + \frac{h}{2}, y_i + \frac{k_2}{2}) \text{ \& } k_4 = hf(x_i + h, y_i + k_3)$$

Start from $y_1 = y_0 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$

with $k_1 = hf(x_0, y_0)$, $k_2 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2})$

$$k_3 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}) \text{ \& } k_4 = hf(x_0 + h, y_0 + k_3)$$

Fourth Order Runge-Kutta Method :-

(a) To solve $\frac{dy}{dx} = f(x, y)$ with the initial condition $y(x_0) = y_0$. let h be the length of intervals for the values of x , then the first increment in y is computed from the formula

$$k_1 = h f(x_0, y_0)$$

$$k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right)$$

$$k_4 = h f(x_0 + h, y_0 + k_3)$$

$$\Delta y = \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

Then $x_1 = x_0 + h$ and $y_1 = y_0 + \Delta y$.

Similarly, we can find the increment for next interval by replacing x_0, y_0 by x_1, y_1 .

(b) Sol. of simultaneous eqns

$$\frac{dy}{dx} = f_1(x, y, z), \quad \frac{dz}{dx} = f_2(x, y, z); \quad y(x_0) = y_0, \quad z(x_0) = z_0$$

Increments k and l in y and z are given by

$$k_1 = h f_1(x_0, y_0, z_0)$$

$$l_1 = h f_2(x_0, y_0, z_0)$$

etc.

(c) Second Order diff eqn: — $\frac{d^2y}{dx^2} = \phi(x, y, \frac{dy}{dx})$

with $y(x_0) = y_0, \quad y'(x_0) = y'_0$.

let $\frac{dy}{dx} = z$. Then $\frac{dz}{dx} = \phi(x, y, z)$

with initial conditions $y(x_0) = y_0, \quad z(x_0) = z_0$.

Ex.1. Given $\frac{dy}{dx} = 1+xy$; $y(0) = 2$. Using Runge-Kutta fourth^{order} method, find $y(0.1)$, $y(0.2)$

Solu: Here $f(x,y) = 1+xy$
 $x_0 = 0$, $y_0 = 2$, $h = 0.1$

$$k_1 = h (1 + x_0 y_0) = .1$$

$$k_2 = .1 f(.05, 2 + .05) = .1 (1 + .05 \times 2.05) = 0.11025$$

$$k_3 = .1 f(.05, 2 + \frac{.11025}{2}) = .1 (1 + .05 \times 2.055125) = 0.110275$$

$$k_4 = .1 f(.1, 2.110275) = .1 (1 + .1 \times 2.110275) = .1 (1.2110275) = .121103$$

$$\therefore \Delta y = \frac{1}{6} (.1 + .22050 + .22055 + .121103) = 0.1103421$$

$$\therefore y(0.1) = y_0 + \Delta y = 2 + 0.1103421 = 2.1103421$$

$$\boxed{y(0.1) = 2.1103421}$$

To calculate $y(0.2)$: $x_1 = .1$, $y_1 = 2.11034$

$$k_1 = .1 (1 + x_1 y_1) = .1 (1 + .211034) = .1211034$$

$$k_2 = .1 f(.1 + .05, 2.11034 + .06055) = .1 f(.15, 2.17089) = 0.13256$$

$$k_3 = .1 f(.15, 2.11034 + .06628) = .1 (1 + .15 \times 2.17662) = .132649$$

$$k_4 = .1 f(.1 + .1, 2.11034 + .132649) = .1 (1 + .2 \times 2.242989) = .144859$$

$$\begin{aligned}\Delta y &= \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \\ &= \frac{1}{6} (.1211034 + 2 \times .13256 + 2 \times .132649 + .144859) \\ &= \frac{1}{6} (.1211034 + .26512 + .265298 + .144859) \\ &= 0.13273\end{aligned}$$

$$\therefore y(0.2) = 2.11034 + 0.13273 = 2.24307$$

Ans. $y(0.1) = 2.11034, y(0.2) = 2.24307$

Ex.2. Use Runge-Kutta method of fourth order to find the numerical solution at $x=1.4$ for $\frac{dy}{dx} = y^2 + x^2$, $y(1) = 0$. Assume step size $h = 0.2$.

Solu: Here $x_0 = 1, y_0 = 0, h = 0.2$.

$$f(x, y) = y^2 + x^2$$

$$k_1 = h f(x_0, y_0) = .2 \times 1 = \boxed{.2}$$

$$k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = h f(1.1, .1)$$

$$= .2 ((1.1)^2 + (.1)^2) = .2 (1.21 + .01) = .244$$

$$k_3 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = .2 f(1.1, .122)$$

$$= .2 (1.21 + .014884) = \cancel{.24578} = .244976$$

$$k_4 = h f(1.2, .244976) = .2 \times (1.44 + .0874858)$$

$$= .3000026$$

$$\therefore \Delta y = \frac{1}{6} (.2 + .488 + \cancel{.59756} + .3000026)$$

$$= \cancel{0.264776} = .246326$$

$$y(1.2) = y_0 + \Delta y = \cancel{.244776} = .246326$$

To calculate $y(1.4)$:

$$x_0 + h = x_1 = 1.2, \quad y_1 = .246326, \quad h = .2$$

$$k_1 = h f(x_1, y_1) = .2 \times (1.2^2 + (.246326)^2) = .2 \times (1.44 + .060676) \\ = .3001352$$

$$k_2 = h f(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}) = .2 \left((1.3)^2 + (.396393)^2 \right) \\ = .3694256$$

$$k_3 = h f(x_1 + h, y_1 + k_2) = .2 (1.4^2 + (.4310388)^2) \\ = .3751589$$

$$k_4 = h f(x_1 + h, y_1 + k_3) = .2 \left((1.4)^2 + (\cancel{.586243})^2 \right) \\ = \cancel{0.5418367} + .4692487$$

$$\therefore \Delta y = \frac{1}{6} (.3001352 + .7388512 + .7503178 + \cancel{.418367}) \\ = \cancel{0.388523483} \\ = 0.376628283$$

$$\therefore y(1.4) = y_1 + \Delta y = .246326 + .376628283 \\ = 0.622954283$$

$$\therefore \boxed{y(1.4) = 0.622954}$$

Ans.

Ex.3. Solve $\frac{dy}{dx} = yz+x$, $\frac{dz}{dx} = xz+y$; given that $y(0) = 1$, $z(0) = -1$ for $y(0.1)$, $z(0.1)$

Soln: — Here $f_1(x, y, z) = yz+x$, $f_2(x, y, z) = xz+y$

$$h = 0.1, \quad x_0 = 0, \quad y_0 = 1, \quad z_0 = -1$$

$$k_1 = hf_1(x_0, y_0, z_0) = h(y_0 z_0 + x_0) = -0.1$$

$$l_1 = hf_2(x_0, y_0, z_0) = h(x_0 z_0 + y_0) = 1(1) = 0.1$$

$$k_2 = hf_1\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) = -0.08525$$

$$l_2 = hf_2\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) = 0.09025$$

$$k_3 = hf_1\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right) = -0.0864173$$

$$l_3 = hf_2\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right) = 0.09096325$$

$$k_4 = hf_1(x_0 + h, y_0 + k_3, z_0 + l_3) = -0.073048$$

$$l_4 = hf_2(x_0 + h, y_0 + k_3, z_0 + l_3) = 0.0822679$$

$$\therefore k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = -0.0860637$$

$$\& \quad l = \frac{1}{6}(l_1 + 2l_2 + 2l_3 + l_4) = 0.0907823$$

$$\text{Thus } y_1 = y(0.1) = y_0 + k = 1 - 0.0860637 = 0.9139363$$

$$z_1 = z(0.1) = z_0 + l = -1 + 0.0907823 = -0.9092176$$

Ans.

$$\boxed{\begin{array}{l} y(0.1) = 0.9139363 \\ z(0.1) = -0.9092176 \end{array}}$$

Predictor - corrector Methods -

In Euler's method

$$y_1 = y_0 + h f(x_0, y_0) \quad \text{--- (1)}$$

$$y_1^{(1)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_0+h, y_1)] \quad \text{--- (2)}$$

① is called predictor ② is called corrector.

In the predictor-corrector method,

four prior values are required for finding the value of y at x_i .

① Milne's ^{P-C} method:

Given $dy/dx = f(x, y)$ and $y = y_0, x = x_0$
to find an approximate value of y for $x = x_0 + h$

The value $y_0 = y(x_0)$ being given, we compute

$$y_1 = y(x_0+h), \quad y_2 = y(x_0+2h), \quad y_3 = y(x_0+3h)$$

by Picard's or Taylor's series method.

Next we calculate

$$f_0 = f(x_0, y_0), \quad f_1 = f(x_0+h, y_1), \quad f_2 = f(x_0+2h, y_2)$$

$$f_3 = f(x_0+3h, y_3)$$

Then we find y_4 which is given by

$$y_4 = y_0 + \frac{4h}{3} (2f_1 - f_2 + 2f_3)$$

which is called a predictor.

Having found y_4 , we obtain a first approx. to

$$f_4 = f(x_0+4h, y_4)$$

Then a better value of y_4 is found as

$$y_4 = y_2 + \frac{h}{3} (f_2 + 4f_3 + f_4)$$

which is called a corrector.

Then an improved value of f_4

is computed and again the corrector is applied ⁽²⁾ to find a still better value of y_4 . We repeat this step until y_4 remains unchanged.

Once y_4 and f_4 are obtained to desired degree of accuracy, $y_5 = y(x_0 + 5h)$ is found from the predictor as

$$y_5 = y_1 + \frac{4h}{3} (2f_2 - f_3 + 2f_4)$$

and $f_5 = f(x_0 + 5h, y_5)$ is calculated. Then a better approximation to the value of y_5 is obtained from the corrector as

$$y_5 = y_3 + \frac{h}{3} (f_3 + 4f_4 + f_5)$$

We repeat this step till y_5 becomes stationary and we then proceed to calculate y_6 as before.

Ex. - Apply Milne's method, to find a soln. of the diff eqn $y' = x - y^2$ in the range $0 \leq x \leq 1$ for the boundary condition $y = 0$ at $x = 0$.

Soln. - Using Picards method, first appx. y_1 is

$$\text{given } h_1 \quad y_1 = y_0 + \int_0^x f(x, y_0) dx = 0 + \int_0^x x dx = \frac{x^2}{2}$$

$$y_2 = \int_0^x (x - \frac{x^4}{2}) dx = \frac{x^2}{2} - \frac{x^5}{20}$$

$$y_3 = \frac{x^2}{2} - \frac{x^5}{20} + \frac{x^8}{160} - \frac{x^{11}}{4400}$$

Now let us take $h = 0.2$. Then

$$x_0 = 0.0$$

$$y_0 = 0.0000$$

$$f_0 = 0.0000$$

$$x_1 = 0.2$$

$$y_1 = 0.0000$$

$$f_1 = 0.1996$$

$$x_2 = 0.4$$

$$y_2 = 0.0795$$

$$f_2 = 0.3932$$

$$x_3 = 0.6$$

$$y_3 = 0.1262$$

$$f_3 = 0.5689$$

Using the predictor $y_4 = y_0 + \frac{4h}{3} (2f_1 - f_2 + 2f_3)$

$$x = 0.8, \quad y_4 = 0.3049, \quad f_4 = 0.7070.$$

and the corrector $y_4 = y_2 + \frac{h}{3} (f_2 + 4f_3 + f_4)$ gives

$$y_4 = 0.3046 \quad \text{--- (2)} \quad f_4 = 0.722$$

Again using the corrector,

$$y_4 = 0.3046 \quad \text{--- (3)}$$

which is same as in (2)

Now using the predictor,

$$y_5 = y_1 + \frac{4h}{3} (2f_2 - f_3 + 2f_4),$$

$$x = 1.0, \quad y_5 = 0.4554, \quad f_5 = 0.7926$$

and the corrector $y_5 = y_3 + \frac{h}{3} (f_3 + 4f_4 + f_5)$ gives

$$y_5 = 0.4555, \quad f_5 = 0.7925$$

Again using the corrector

$$y_5 = 0.4555, \quad \text{a value which is same as } y_5$$

$$\text{Hence } y(1) = 0.4555.$$

R

⑥ Adams - Bashforth method:

Given $\frac{dy}{dx} = f(x, y)$ and $y_0 = y(x_0)$

we compute

$$y_{-1} = y(x_0 - h), y_{-2} = y(x_0 - 2h), y_{-3} = y(x_0 - 3h) \text{ by}$$

Taylor's series or Runge-Kutta method.

Next we calculate

$$f_{-1} = f(x_0 - h, y_{-1}), f_{-2} = f(x_0 - 2h, y_{-2}), f_{-3} = f(x_0 - 3h, y_{-3}).$$

Then to find y_1 , we use

$$y_1 = y_0 + \frac{h}{24} (55f_0 - 59f_{-1} + 37f_{-2} - 9f_{-3})$$

which is called Adams - Bashforth predictor formula.

Having found y_1 , we find $f_1 = f(x_0 + h, y_1)$

Then to find a better value of y_1 , we use

$$y_1 = y_0 + \frac{h}{24} (9f_1 + 19f_0 - 5f_{-1} + f_{-2})$$

which is called Adams - Moulton corrector formula.

Then an improved value of f_1 is calculated and again the corrector is applied to find a still better value y_1 . This step is repeated till y_1 remains unchanged and then we proceed to calculate y_2 as above.

Ex. Solve the initial value problem $\frac{dy}{dx} = x - y^2$
 $y(0) = 1$ to find $y(0.4)$ by Adams method.
 Starting values required are to be obtained by Runge-Kutta method of order 4 using step value $h = 0.1$.

Soln.

We have $f(x, y) = x - y^2$

To find $y(0.1)$.

Here $x_0 = 0$, $y_0 = 1$ $h = 0.1$

$$\therefore k_1 = -0.1000, k_2 =$$

$$y(0.1) = y_1 = y_0 + k_1 = 1 - 0.0883 = 0.9117.$$

$$y(0.2) = 0.8494$$

$$y(0.3) = 0.8061.$$

Now the starting values of Adams' method with $h=0.1$ are :

$$x = 0.0 \quad y_{-3} = 1.0000 \quad f_{-3} = 0.0 - (1.0)^2 = -1.0000$$

$$x = 0.1 \quad y_{-2} = 0.9117 \quad f_{-2} = 0.1 - (0.9117)^2 = -0.7312$$

$$x = 0.2 \quad y_{-1} = 0.8494 \quad f_{-1} = 0.2 - (0.8494)^2 = -0.5215$$

$$x = 0.3 \quad y_0 = 0.8061 \quad f_0 = 0.3 - (0.8061)^2 = -0.3498$$

Using the predictor,

$$y_1 = y_0 + \frac{h}{24} (55f_0 - 59f_{-1} + 37f_{-2} - 9f_{-3})$$

$$\begin{aligned} x = 0.4, \quad y_1 &= 0.8061 + \frac{0.1}{24} (55(-0.3498) - 59(-0.5215) \\ &\quad + 37(-0.7312) - 9(-1)) \\ &= 0.7789. \end{aligned}$$

$$\therefore f_1 = -0.2067$$

Using the corrector,

$$y_1 = y_0 + \frac{h}{24} [9f_1 + 19f_0 - 5f_{-1} + f_{-2}]$$

$$\begin{aligned} y_1 &= 0.8061 + \frac{0.1}{24} [9(-0.2067) + 19(-0.3498) - 5(-0.5215) \\ &\quad - 0.7312] \\ &= 0.7785 \end{aligned}$$

$$\text{Hence } y(0.4) = 0.7785$$

Multistep methods:

For multistep method we need more than 1 values of y .

Predictor-corrector methods:

i) Milne's method: Solve $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$.
To find y_i , we need y_{i-1}, y_{i-2}

y_{i-3} & y_{i-4} .

i.e. to find y_4 , we must have y_0, y_1, y_2, y_3 .

To find y_1, y_2, y_3 we use R.K. method

Predictor: $y_4 = y_0 + \frac{4h}{3} (2f_1 + f_2 + 2f_3)$

Corrector: $y_4 = y_2 + \frac{h}{3} (f_2 + 4f_3 + f_4)$

compute f_4 using corrector.

use this f_4 in corrector. Find f_4 repeat the process until two consecutive values of y_4 are repeated.

EX. Using Milne's method, solve

$$\frac{dy}{dx} = xy + y^2, \quad y(0) = 1$$

for $x = 0.4$

Soln: We find $y(0.1)$ $y(0.2)$ $y(0.3)$ by

R.K. method.

(i) To find $y(0.1)$

Here $x_0 = 0$, $y_0 = 1$, $h = 0.1$

$$K_1 = hf(x_0, y_0) = 0.1000$$

$$K_2 = 0.1155$$

$$K_3 = 0.1172$$

$$K_4 = 0.13598$$

$$\therefore y_1 = y(0.1) = 1 + \frac{1}{6}(0.1 + 0.231 + 0.2343 + 0.1359) \\ = \boxed{1.1169}$$

To find $y(0.2)$:

$$x_1 = 0.1, y_1 = 1.1169 \quad h = 0.1$$

$$k_1 = 0.1359$$

$$k_2 = 0.1581$$

$$k_3 = 0.1609$$

$$k_4 = 0.1888$$

$$\therefore y_2 = y(0.2) = 1.1169 + \frac{1}{6}(0.1359 + 2 \times 0.1581 + 2 \times 0.1609 + 0.1888) \\ = \boxed{1.2773}$$

To find $y(0.3)$:

$$k_1 = 0.1887, k_2 = 0.2224, k_3 = 0.2275, k_4 = 0.2726$$

$$\therefore y(0.3) = y_2 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = \boxed{1.509}$$

Using the predictor $y_4 = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3)$

$$x_4 = 0.4, y_4 = 1.8344, f_4 = 4.0988$$

and the corrector, $y_4 = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4)$ yields

$$y_4 = 1.2773 + \frac{0.1}{3}[1.8869 + 4(2.7132) + 4.0988]$$

$$= 1.8386$$

$$f_4 = 4.1159$$

Again using corrector :

$$y_4 = 1.2773 + \frac{0.1}{3}[1.8869 + 4(2.7132) + 4.1159]$$

$$= 1.8391$$

$$f_4 = 4.1182$$

Again using corrector

$$y_4 = 1.2773 + \frac{0.1}{3}[1.8869 + 4(2.7132) + 4.1182]$$

$= 1.8392$ which is same as above.

$$\text{Hence } y(0.4) = 1.8392$$

3. Adams-Bashforth method:

$$\frac{dy}{dx} = f(x, y); y_0 = y(x_0)$$

We compute three values of y viz y_{-1}, y_{-2}, y_{-3} to find y_1

$$\text{Predictor: } y_1 = y_0 + \frac{h}{24} (55f_0 - 59f_{-1} + 37f_{-2} - 9f_{-3})$$

$$\text{Corrector: } y_1 = y_0 + \frac{h}{24} (9f_1 + 19f_0 - 5f_{-1} + f_{-2})$$

An improved value of f_1 is obtained & again using corrector we get y_1 to get better value of f_1

Ex. Given $\frac{dy}{dx} = x^2(1+y)$ and $y(1) = 1, y(1.1) = 1.233$

$y(1.2) = 1.548, y(1.3) = 1.979$, evaluate $y(1.4)$ by

Adams-Bashforth method.

Soln: $f(x, y) = x^2(1+y)$

Starting values of the Adams-Bashforth method with

$h = 0.1$ are

$$x = 1.0$$

$$y_{-3} = 1.000$$

$$f_{-3} = (1.0)^2(1+1.000) = 2.000$$

$$x = 1.1$$

$$y_{-2} = 1.233$$

$$f_{-2} = 2.702$$

$$x = 1.2$$

$$y_{-1} = 1.548$$

$$f_{-1} = 3.669$$

$$x = 1.3$$

$$y_0 = 1.979$$

$$f_0 = 5.035$$

Using the predictor: $y_1 = y_0 + \frac{h}{24} (55f_0 - 59f_{-1} + 37f_{-2} - 9f_{-3})$

$$x_1 = 1.4, y_1 = 2.573, f_1 = 7.004$$

Using the corrector:

$$y_1 = y_0 + \frac{h}{24} (9f_1 + 19f_0 - 5f_{-1} + f_{-2})$$

$$= 1.979 + \frac{0.1}{24} (9 \times 7.004 + 19 \times 5.035 - 5 \times 3.669 + 2.702)$$

$$= 2.575$$

Simultaneous first order diff. eqns.

The simultaneous diff eqns

of the type

$$\frac{dy}{dx} = f(x, y, z) \quad \text{--- (1)}$$

$$\text{and } \frac{dz}{dx} = \phi(x, y, z) \quad \text{--- (2)}$$

with initial conditions $y(x_0) = y_0$ and $z(x_0) = z_0$ can be solved by the methods discussed in the preceding sections, especially Picard or Runge-Kutta methods:

(i) Picard's method gives

$$y_1 = y_0 + \int_{x_0}^{x_1} f(x, y_0, z_0) dx, \quad z_1 = z_0 + \int_{x_0}^{x_1} \phi(x, y_0, z_0) dx$$

$$y_2 = y_0 + \int_{x_0}^{x_2} f(x, y_1, z_1) dx, \quad z_2 = z_0 + \int_{x_0}^{x_2} \phi(x, y_1, z_1) dx$$

etc.

(ii) Runge-Kutta method is applied as follows:

Starting at (x_0, y_0, z_0) and taking the step size

to be h , we calculate

$$l_1 = h \phi(x_0, y_0, z_0)$$

$$k_1 = h f(x_0, y_0, z_0)$$

$$k_2 = h f(x_0 + \frac{h}{2}, y_0 + \frac{1}{2}k_1, z_0 + \frac{1}{2}l_1)$$

$$l_2 = h \phi(x_0 + \frac{h}{2}, y_0 + \frac{1}{2}k_1, z_0 + \frac{1}{2}l_1)$$

$$k_3 = h f(x_0 + \frac{h}{2}, y_0 + \frac{1}{2}k_2, z_0 + \frac{1}{2}l_2)$$

$$l_3 = h \phi(x_0 + \frac{h}{2}, y_0 + \frac{1}{2}k_2, z_0 + \frac{1}{2}l_2)$$

$$k_4 = h f(x_0 + h, y_0 + k_3, z_0 + l_3)$$

$$l_4 = h \phi(x_0 + h, y_0 + k_3, z_0 + l_3)$$

$$\therefore k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$\& \quad l = \frac{1}{6}(l_1 + 2l_2 + 2l_3 + l_4)$$

$$\text{Hence } \left. \begin{aligned} y_1 &= y_0 + k \\ z_1 &= z_0 + l \end{aligned} \right\}$$

To compute y_2 & z_2 , we simply replace x_0, y_0, z_0 by x_1, y_1, z_1 in the above formulae.

Ex. Using Picards method, find appropriate values of y and z corresponding to $x=0.1$, given that $y(0)=2, z(0)=1$ and

$$\frac{dy}{dx} = x+y, \quad \frac{dz}{dx} = x-y^2$$

~~$$y(0)=2, \quad z(0)=1$$~~

Sol. Here $x_0 = 0, y_0 = 2, z_0 = 1$

$$f(x, y, z) = x+y$$

$$\phi(x, y, z) = x-y^2$$

$$\therefore \text{First appr. } y_1 = y_0 + \int_{x_0}^x f(x, y_0, z_0) dx$$

$$= 2 + \int_0^x (x+1) dx = 2 + x + \frac{x^2}{2}$$

$$z_1 = z_0 + \int_{x_0}^x \phi(x, y_0, z_0) dx$$

$$= 1 + \int_0^x (x-4) dx = 1 - 4x + \frac{1}{2}x^2$$

2nd appr.

$$y_2 = y_0 + \int_{x_0}^x f(x, y_1, z_1) dx$$

$$= 2 + \int_0^x (x + 1 - 4x + \frac{1}{2}x^2) dx = 2 + x - \frac{3}{2}x^2 + \frac{x^3}{6}$$

$$z_2 = z_0 + \int_{x_0}^x \phi(x, y_1, z_1) dx = 1 + \int_0^x [x - (2 + x - \frac{3}{2}x^2 + \frac{x^3}{6})^2] dx$$

$$= 1 - 4x + \frac{3}{2}x^2 - x^3 - \frac{x^4}{4} - \frac{x^5}{20}$$

$$= 1 - 4x + \frac{3}{2}x^2 - x^3 - \frac{x^4}{4} - \frac{x^5}{20}$$

Third appr.

$$y_3 = y_0 + \int_{x_0}^x f(x, y_2, z_2) dx$$

$$= 2 + x - \frac{3}{2}x^2 - \frac{1}{2}x^3 - \frac{1}{4}x^4 - \frac{1}{20}x^5 - \frac{1}{120}x^6$$

$$z_3 = z_0 + \int_{x_0}^x \phi(x, y_2, z_2) dx$$

$$= 2 - 4x - \frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{7}{12}x^4 - \frac{31}{60}x^5 + \frac{1}{12}x^6 - \frac{1}{252}x^7$$

and so on

when $x = 0.1$

$$y_1 = 2.105$$

$$y_2 = 2.08517$$

$$y_3 = 2.08447$$

$$z_1 = 0.605$$

$$z_2 = 0.58397$$

$$z_3 = 0.58672$$

Hence $y(0.1) = 2.0845$, $z(0.1) = 0.5867$.

correct to four decimal places.

Second order diff. eqns:-

$$\text{let } \frac{d^2y}{dx^2} = f(x, y, \frac{dy}{dx})$$

with initial conditions $y(x_0) = y_0$ $y'(x_0) = y_1$.

Thus by writing $dy/dx = z$, it can be reduced to two first order simultaneous diff eqns

$$\frac{dy}{dx} = z, \quad y(x_0) = y_0, \quad \frac{dz}{dx} = f(x, y, z), \quad z(x_0) = y_1$$

These eqns can be solved as explained above.

Ex. Using Runge-Kutta method, solve, $y'' = 2y^{1/2} - y$
for $x = 0.2$ correct to 4 decimal places. Initial conditions are

$$x = 0, y = 1, y' = 0.$$

Solution: let $\frac{dy}{dx} = z = f(x, y, z)$

$$\text{Then } \frac{dz}{dx} = 2z^{1/2} - y = \phi(x, y, z)$$

$$\text{Here } x_0 = 0, y_0 = 1, z_0 = 0, h = 0.2$$

Using Runge-Kutta method

$$k_1 = h f(x_0, y_0, z_0) \\ = 0.2 \times 0 = 0$$

$$l_1 = h \phi(x_0, y_0, z_0) \\ = 0.2 \times (-1) = -0.2$$

$$k_2 = h f(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}) \\ = -0.02$$

$$k_2 = h f(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}) \\ = -0.1998$$

$$k_3 = -0.02$$

$$l_3 = -0.1970$$

$$k_4 = -0.0392$$

$$l_4 = -0.1905$$

$$k = -0.0199$$

$$l = -0.1970$$

Hence at $x = 0.2$

$$y = y_0 + k = 1 - 0.0199 = 0.9801$$

2

$$y' = z = z_0 + l = 0 - 0.1970 = -0.1970$$

} Ans.

* Second order differential equations:

$$\frac{d^2 y}{dx^2} = f(x, y, \frac{dy}{dx})$$

By writing $dy/dx = z$, it can be reduced to two first order simultaneous differential equations

$$\frac{dy}{dx} = z, \quad \frac{dz}{dx} = f(x, y, z)$$

and can be solved.

Ex. Using Runge-Kutta fourth order method, solve
 $y'' = xy^2 - y^2$ for $x = 0.2$ correct to four decimal places. Initial conditions are $x=0, y=1, y'=0$.

Soln: Let $dy/dx = z = f(x, y, z)$

Then $\frac{dz}{dx} = xz^2 - y^2 = \phi(x, y, z)$

We have $x_0 = 0, y_0 = 1, z_0 = 0$. Take $h = 0.2$

$\therefore$ From R.K. method

$$k_1 = hf(x_0, y_0, z_0) = z_0 = 0$$

$$k_2 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{h}{2}k_1)$$

$$= 0.2[-\frac{1}{2}(0.2)] = -0.02$$

$$k_3 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{h}{2}k_2)$$

$$= 0.2(-0.0999) = -0.02$$

$$k_4 = hf(x_0 + h, y_0 + k_3, z_0 + k_3)$$

$$= 0.2(-0.1958) = -0.0392$$

$$\therefore k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$= -0.0199$$

Hence at $x = 0.2$

$$y = y_0 + k = 1 - 0.0199 = 0.9801$$

$$y' = z = z_0 + l = 0 - 0.1970 = -0.1970$$

Ans

$$\begin{aligned} l_1 &= h\phi(x_0, y_0, z_0) = 0.2[0.0 - 1] = -0.2 \\ l_2 &= h\phi(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{h}{2}l_1) \\ &= 0.2[-0.999] = -0.1998 \\ l_3 &= h\phi(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{h}{2}l_2) \\ &= 0.2(-0.9791) = -0.1958 \\ l_4 &= h\phi(x_0 + h, y_0 + k_3, z_0 + k_3) \\ &= 0.2(-0.9527) = -0.1905 \\ l &= \frac{1}{6}(l_1 + 2l_2 + 2l_3 + l_4) \\ &= -0.1970 \end{aligned}$$

Numerical Solution of Partial Differential Equations

Classification of second order equations

The general linear partial differential equation of the second order in two independent variables is of the form

$$A(x,y) \frac{\partial^2 u}{\partial x^2} + B(x,y) \frac{\partial^2 u}{\partial x \partial y} + C(x,y) \frac{\partial^2 u}{\partial y^2} + f(x,y,u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}) = 0 \quad \text{--- (1)}$$

or $A(x,y)u_{xx} + B(x,y)u_{xy} + C(x,y)u_{yy} + f(x,y,u,u_x,u_y) = 0$
Such a partial differential equation is said to be

i) elliptic if $B^2 - 4AC < 0$

ii) parabolic if $B^2 - 4AC = 0$

iii) hyperbolic if $B^2 - 4AC > 0$

Symbols:

$$\frac{\partial^2 u}{\partial x^2} = u_{xx}, \quad \frac{\partial^2 u}{\partial y^2} = u_{yy}$$

$$\frac{\partial^2 u}{\partial x \partial y} = u_{xy}$$

$$\frac{\partial u}{\partial x} = u_x, \quad \frac{\partial u}{\partial y} = u_y$$

Ex. 1. (i) classify the following equations:

i) $\frac{\partial^2 u}{\partial x^2} + 4 \frac{\partial^2 u}{\partial x \partial y} + 4 \frac{\partial^2 u}{\partial y^2} - \frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} = 0$

ii) $x \frac{\partial^2 u}{\partial x^2} + (1-y^2) \frac{\partial^2 u}{\partial y^2} = 0, \quad -\infty < x < \infty; \quad -1 < y < 1$

iii) $(1+x^2) \frac{\partial^2 u}{\partial x^2} + (5+2x^2) \frac{\partial^2 u}{\partial x \partial y} + (4+x^2) \frac{\partial^2 u}{\partial y^2} = 0$

Soln: i) $A = 1, B = 4, C = 4$

$$B^2 - 4AC = 16 - 4 \cdot 1 \cdot 4 = 0$$

$\therefore$ Eqn (i) is parabolic.

ii) Here $A = x^2, B = 1-y^2, C = 0$

$$B^2 - 4AC = 0 - 4 \cdot x^2 \cdot (1-y^2) = -4x^2(1-y^2) = 4x^2(y^2-1)$$

For all x between $-\infty$ to ∞ , x^2 is positive

For all y between -1 to 1 , y^2 is less than 1

$$\therefore B^2 - 4AC = 4x^2(y^2-1) < 0.$$

$\therefore$ Eqn (ii) is elliptic.

iii) Here, $A = 1+x^2, B = 5+2x^2, C = 4+x^2$

$$\therefore B^2 - 4AC = (5+2x^2)^2 - 4(1+x^2)(4+x^2) = 9 > 0$$

$\therefore$ Eqn (iii) is hyperbolic.